

Neonatal and Pediatrics Anesthesia

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Presentation outline

- Introduction
- The transition period
- The systemic overview of the pediatric patients
 - ✓CVS
 - ✓RS
 - ✓Thermoregulation
 - ✓Fluid management
 - ✓Pain management
 - ✓Equipments
 - ✓POP Concerns
 - ✓Emergence delirium
 - ✓Child preparation
 - ✓Neonatal resuscitation

INTRODUCTION

- Pediatric patients are not small adults.
- Neonates are from 0–1 months
- Infants are from 1–12 months
- Toddlers are from 1–3 years
- Small children are from 4–12 years of age, they all differing anesthetic requirements

CONT'D

- Knowledge of physiological, anatomic, pharmacological, and psychological difference is important for safe anesthesia for each group
- Modification of anesthetic equipment and techniques are needed
- Morbidity and mortality risk is generally inversely proportional to age, neonates being at highest risk
- In addition, pediatric patients are prone to illnesses that require unique surgical and anesthetic strategies

Physiological

- Heart-rate-dependent cardiac output
- Faster heart rate
- Lower blood pressure
- Faster respiratory rate
- Lower lung compliance
- Greater chest wall compliance

CONT'D

- Lower functional residual capacity
- Higher ratio of body surface area to body weight
- Higher total body water content
- Short pain threshold
- Prone to hypothermia or MH
- Minimal glycogen stores, neonate's tend to have hypoglycemia and acidemia

Anatomic

- Noncompliant left ventricle
- Residual fetal circulation
- Difficult venous and arterial cannulation
- Large head and tongue
- Narrow nasal passages

CONT'D

- Anterior and cephalad larynx
- Long epiglottis
- Short trachea and neck
- Prominent adenoids and tonsils
- Weak intercostal and diaphragmatic muscles
- High resistance to airflow

Pharmacological

- Immature hepatic biotransformation
- Decreased protein binding
- Rapid rise in FA/FI-- rapid inhalational induction and desaturation
- Rapid induction and recovery
- Increased minimum alveolar concentration

CONT'D

- Larger volume of distribution for water-soluble drugs
- Immature neuromuscular junction
- Reduced ability to excrete free water and solute loads
- In infants and neonates much less is known about the pharmacodynamics of anesthetic agents

Psychological

- There is association between preoperative anxiety and adverse postoperative outcomes like:
 - ✓ Emergence delirium
 - ✓ Increased analgesic requirement
 - ✓ Negative behavioral changes

CONT'D

- In general, fear in children can be related to a number of factors:
 - ✓ Parental separation
 - ✓ Unfamiliar and threatening hospital environment
 - ✓ Painful procedures
 - ✓ The operation itself, and anesthesia

CONT'D

- Infants up to 9 months are less anxious to separation
- Separation anxiety is the greatest problem in children from 1-3 years
- Different distraction techniques
- Parents and OR staff should be involved in perioperative plan and management of aggressive combatant children

CONT'D

- For very anxious children, intranasal premedication is beneficial
- If no IV line, IM ketamine (4-5 mg/kg) or
- Inhalational induction using high concentrations of sevoflurane can be a helpful option to induce anesthesia in this population

INFANT PHYSIOLOGY AND TRANSITION PERIOD

- Before birth, fetal growth and development depend on the:
 - ✓ Fetal genetic composition
 - ✓ Mother's placental function, and
 - ✓ Potential exposure to chemicals or infectious agents
- After birth, the newborn rapidly adjusts to the extrauterine environment
- The changes in functions of systems will determine viability of the neonate, as well as its ability to grow and develop properly

CVS

- At birth the circulation undergoes a basic change as blood oxygenation occurs through the lungs (not placenta)
- Less compliant ventricles
- They are more dependent on HR for CO
- Prone to myocardial depression by potent anesthetics (less cardiac Ca^{++} store)
- They are dependent on ionized blood calcium (exogenous)

CONT'D

- Children are at risk for:
 - ✓ Cardiac arrhythmias
 - ✓ Hyperkalemia
 - ✓ Rhabdomyolysis
 - ✓ Myoglobinemia
 - ✓ Masseter spasm, and
 - ✓ Malignant hyperthermia (MH) after succinylcholine admin.
- Atropine pre-treatment prevents hypotension and bradycardia after Succ

Respiratory System

- The neonatal airway differs from the adult airway in:
 - ✓ Large tongue size (airway obstruction and intubation difficulty at induction)
 - ✓ More cephalic (anterior) larynx straight blades are more preferred
 - ✓ Short and omega shaped epiglottis angled over the laryngeal inlet
 - ✓ Angled vocal cords (blindly passed ETT may easily lodge in the esophagus)
 - ✓ Funnel shaped larynx (**cricoid cartilage** is the narrowest portion until 5year)

CONT'D

- Neonates and infants have:
 - ✓ Larger head and tongue
 - ✓ Narrow nasal passages
 - ✓ Anterior and cephalad larynx
 - ✓ Long epiglottis and
 - ✓ Short trachea and neck

CONT'D

- Hence the neonates and young infants are obligate nasal breathers until 5 months of age
- Confirm contralateral nare patency to avoid complete blockade of AW

CONT'D

- Pediatrics (upto 13yrs) have relatively **↑ alveolar ventilation and ↓ FRC** compared with adolescents (13-18 yrs) and adults
- This higher MV -to-FRC ratio **speeds up inhalational induction**
- MAC is higher in infants than in neonates and adults for IAAs
- But, **sevoflurane** has the same MAC in neonates and infants
- It also have wider therapeutic index than halothane and has become a preferred induction agent in pediatric anesthesia

PHARMACOLOGIC

- Neonates have ↑ volumes of distribution and ↓ clearances for drugs
- They need larger loading doses and continuous infusion rates or dose intervals tend to be longer
- Allometric scaling (e.g., body mass) can predict the dose requirement for most drugs in children better than simple mg/kg calculations

THERMOREGULATION

- Neonates have thin skin, low fat content, and a higher surface relative to weight
- They easily lose heat to the environment
- This heat loss can be enhanced by
 - ✓ Cold operating rooms
 - ✓ wound exposure
 - ✓ IV fluids administration
 - ✓ Dry anesthetic gases, and
 - ✓ The direct effect of anesthetic agents on temperature regulation

CONT'D

- They are prone to intraoperative hypothermia
- Hypothermia has been associated with:
 - ✓ Delayed awakening from anesthesia
 - ✓ Cardiac irritability
 - ✓ Respiratory depression
 - ✓ Increased pulmonary vascular resistance
 - ✓ Altered drug responses

CONT'D

- Efforts to maintain a warm surgical unit by using:
 - ✓ Hot air mattresses,
 - ✓ Warm surgical skin preparation solutions, and
 - ✓ Transport of the neonate or infant in a warm device
 - ✓ Keeping the infant covered at transport helps to prevent hypothermia

FLUID MANAGEENT

- Meticulous fluid management is required in pediatrics
- They have an extremely limited margins of error
- A programmable infusion pump should be used for accuracy
- Drugs are flushed through low dead-space tubing to minimize unnecessary fluid administration
- They are also very prone to hyponatremia
- Remedy, Isotonic solutions should be used

CONT'D

- Three important perioperative fluid mgt goals in pediatrics is to:
 1. Meet maintenance requirements
 2. Replace preoperative deficits and
 3. Compensate for ongoing fluid and blood losses
- The most commonly used basic fluid maintenance formula in pediatric is based on relationship b/n physiologic fluid loss and caloric expenditures
- The 4-2-1 formula bases the above principle
- But, the infusion may be initiated with the rate of 10 mL/kg/h and adjusted to actual requirement during the further course(Germany guideline in Europe)
- Depend largely on procedure type and on the pathological state of the child

PAIN MANAGEMENT

- Pediatrics require adequate analgesia for painful procedures
- They have a short pain threshold
- Unclear optimal dose of general anesthetics for adequate analgesia
- The bispectral index (BIS) cannot be used in this age group to guide depth of anesthesia

EQUIPMENTS

- Pediatric anesthesia requires appropriate pediatric equipment in a range of sizes
- Neonates require ventilators that are designed to meet their needs

POP CONCERNS

- Peri-operative apnea and laryngospasm are common
- Laryngospasm can be avoided by extubating the patient either awake or while deeply anesthetized
- Extubation between these extremes is recognized as hazardous

Emergence delirium

- Postoperative delirium / agitation is common for preschool children
- Agitation may be due to pain, fear, hunger, and delirium
- Many strategies are known to reduce the risk of delirium
- Children receiving propofol anesthesia are at lower risk for delirium than those receiving volatile anesthetics

Child preparation

- A viral infection within 2–4 weeks before GA and ETT intubation increased risk for perioperative pulmonary complications like:
 - ✓ Wheezing,
 - ✓ Laryngospasm,
 - ✓ Hypoxemia
 - ✓ Atelectasis

CONT'D

- Preoperative anxiety is common in children, hence:
 - ✓ Different distraction techniques
 - ✓ Premedication with midazolam or $\alpha 2$ agonists
 - ✓ Parental presence at induction have all been shown to reduce anxiety

CONT'D

- Special consideration is needed when planning an anesthetic for a neonate
- Use of caffeine as treatment and prophylaxis Vs POP apnea
- Use of oxygen concentration Vs retinopathy of prematurity
- Use of anesthetic agents Vs neurodevelopmental effects fetal and neonatal brain

CONT'D

- True surgical emergencies are uncommon in the neonatals
- Knowledge of conditions with comorbidities like:
 - ✓ Tracheoesophageal fistula
 - ✓ Omphalocele, and gastroschisis
 - ✓ Congenital diaphragmatic hernia, and a thorough preoperative evaluation and stabilization is mandated

Neonatal Resuscitation

- Profound CVS and RS changes occur at birth
- Failure to make these changes may result in death or CNS injury
- The earlier the resuscitation the better the outcome

INITIAL ASSESSMENT OF THE FETUS AT BIRTH

- Rapid evaluation of the newborn to determine if the infant is:
 - ✓ Term
 - ✓ Breathing normally, or crying
 - ✓ Has a normal tone (see next slide)

Evaluation of the Newborn

Clinical Condition	Intervention
Initial resuscitation	Clear infant airway Warm, dry, stimulate, position Evaluate HR, respirations, color
HR > 100 beats/min, breathing, no cyanosis	Observation
HR > 100 beats/min, but persistent respiratory distress or cyanosis	Clear airway, SpO2 monitoring, Consider CPAP
Apnea, gasping, or HR < 100 beats/min	Bag-mask PPV, SpO2 monitoring
After initiation of resuscitation (PPV), HR > 100 beats/min, effective ventilation	Post-resuscitation care
HR < 60 beats/min	Consider intubation, Chest compressions, Coordinate PPV
HR = 60-100 beats/min	Continue with PPV, SpO2 monitoring

ONGOING ASSESSMENT

- Ongoing assessment consists of three signs:
- Heart rate, respirations, and oxygenation
- Preferably determine the HR is by auscultation
- All of these vital signs should be determined within the first 30s

Perioperative Management

PREOPERATIVE EVALUATION

- It is aimed to evaluate the:
 1. Child's medical conditions
 2. Needs of the planned surgical or diagnostic procedure
 3. Psychological context of the child and family
- If indicated, complete it with specific preoperative tests and specialty consultations

CONT'D

- Effective evaluation, preparation, and anesthetic management depends on our knowledge, clinical skills, and vigilance
- Anesthetic equipment and monitoring devices, airway management, drug choice, fluid management, temperature control, surgical needs, pain management, and postoperative care must be addressed
- The neonates have stress responses similar to those of older patients

Medical history

- Preanesthesia evaluation time depends on patient's status and type of surgery or
- Institutional policy , and demographic and geographic conditions
- Take the details of previous anesthesia experiences, and family history
- Obtaining this information is helpful (absent hospital charts)

CONT'D

- Get much relevant information as possible, and emphasis on:
 - ✓ Maternal factors that may have affected fetal growth
 - ✓ The current status of the newborn
- Previously intubated newborns may have subglottic edema
- More likely are signs of lower airway disease, such as tachypnea, rhonchi, retractions, and cyanosis

CONT'D

- Important factors include the history of:
 - ✓ Feeding and hydration
 - ✓ Need for oxygenation or ventilatory support
 - ✓ Cardiovascular abnormalities
 - ✓ Evidence of CNS problems such as seizures or intraventricular hemorrhage
 - ✓ Gestational status

CONT'D

- This may be related to the early development of RDS or it may be
- Due to Meconium aspiration, pneumonia, pneumothorax, or HF
- The cause of any respiratory distress needs to be evaluated clearly prior to anesthesia to identify treatable causes and begin therapy
- The WHO definition of prematurity as <37 weeks' gestation at birth
- A birth weight of $\leq 2,500$ g are subject to high anesthetic risks

CONT'D

- RDS, intraventricular hemorrhage, sepsis, necrotizing enterocolitis (NEC), bronchopulmonary dysplasia are very high in low birth weight (LBW) infants
- Most common abnormalities found in the preterm population with implications for anesthetic evaluation, preparation, and management.(next slides)

Respiratory

- Respiratory distress syndrome
- Apnea
- Pneumothorax, pneumomediastinum
- Pneumonia
- Pulmonary hemorrhage
- Bronchopulmonary dysplasia

Cardiovascular

- Patent ductus arteriosus (persisted fetal circulation)
- Hypotension
- Bradycardia
- Pulmonary hypertension (commonly from hypoxia)
- Persistent transitional circulation
- Congenital heart disease

Central nervous

- Hypoxic-ischemic encephalopathy
- Seizures
- Kernicterus
- Drug withdrawal

Metabolic

- Hypoglycemia
- Hyperglycemia
- Hypocalcemia
- Hypothermia
- Metabolic acidosis

Renal

- Hyponatremia
- Hypernatremia
- Hyperkalemia
- Poor urine output

Gastrointestinal

- Poor feeding
- Necrotizing enterocolitis
- Intestinal obstruction

Hematologic

- Anemia
- Hyperbilirubinemia
- Vitamin K deficiency

Other

- Retinopathy of prematurity
- Sepsis and infections

Physical Examination (P/E)

- Age specific focused on conditions requiring surgical intervention
- Hydration status is very important (fasting and losses related to the surgical lesion)
- Correct any deficits in pediatrics before surgery, except in extreme, life-threatening emergency situations

CONT'D

- Clinical signs of dehydration includes:
 - Sunken fontanelle
 - Poor skin turgor
 - Dry mucus membranes
 - Sunken eyes
 - Poor skin perfusion
 - Delayed capillary refill
 - Hypothermia
 - History of tachycardia or absent urine output

CONT'D

- Also focus on examining the CVS and respiratory systems
- For CVS evaluate for poor perfusion or pulses, abnormal rhythm or rate, a murmur or gallop, hepatomegaly etc.
- If there is murmur, consulting cardiologist is important for best outcome

CONT'D

- On the respiratory system(RS) evaluate for presence of stridor or airway obstruction, such as sternal or chest wall retractions
- Although upper airway obstruction (UAWO) is uncommon in newborn, laryngeal webs,tongue or supraglottic region cysts, vocal cord paralysis, and hemangiomas of the airway can cause obstruction

CONT'D

- P/E for ASA I and II status patients ↓ses both procedure- and anesthesia-related anxiety
- P/E includes a thorough assessment of :
 - ✓ The airway
 - ✓ The cardiovascular system
 - ✓ The respiratory system
 - ✓ The nervous systems along with the hydration state of the child
- Routine preoperative tests are not usually mandated???
- Order a focused tests to suspected condition (CHD,Asthma...etc) if needed

CONT'D

- Optional neonatal screening for long QT syndrome (LQTS)
- It is the leading suspected cause of sudden infant death
- Pre-op stress of and many anesthetics may prolong the QT
- Otherwise, routines are radiologic and echocardiographic evaluation

A Child With an Upper Respiratory Infection(URI)

- More than 200 viruses have been shown to be associated with URI
- Children commonly get URI 6-8 times a year
- Usually last for 7-10 days, but symptoms may persist for 3wks
- This leads to an inflammatory response that results in airway edema and increased secretions

CONT'D

- Bronchial hyperreactivity may persist for 6-7 weeks or longer, well beyond the disappearance of clinical symptoms
- Hence, a child with a URI is a major challenge for pediatric anesthesia
- Most common associated perioperative respiratory complications:
 - ✓ Wheezing (10-fold)
 - ✓ Laryngospasm (5-fold)
 - ✓ Bronchospasm

CONT'D

- ✓ Breath holding
- ✓ Atelectasis
- ✓ Arterial oxygen desaturation
- ✓ Bacterial pneumonia
- ✓ Unplanned hospital admission

This is particularly likely if the child has a productive cough, high fever, green runny nose, otitis media, or a family history of reactive airway disease

CONT'D

- Elicit any history or signs of primary pulmonary morbidity:
 - ✓Bronchial asthma
 - ✓Prematurity
 - ✓Bronchopulmonary dysplasia
 - ✓Cystic fibrosis
 - ✓Pulmonary hypertension
 - ✓Passive smoke exposure should also be considered

CONT'D

- ENT, and eye as well as upper abdominal and thoracic surgeries ↑ the risk of perioperative lung complications
- Decision to proceed with anesthesia in these children depends on severity of URI symptoms, presence of coexisting illnesses, & urgency of surgery

Mild cold

- Runny nose
- Dry cough
- **Watery congestion**

- **Pretreatment with salbutamol**
- Anticholinergic premedication
- Mask ventilation
- Humidification of inspired gases
- Avoid desflurane
- Use LMA
- Use propofol
- Consider lidocaine
- Longer than usual stay in the recovery room should be planned

Moderate cold

- Runny nose
- Moist cough
- **Purulent congestion**
- Perform Risk-benefit-ratio
- **If the following risk factors are an issue defer for ≥ 2 weeks**
 - ✓ Child < 1 year
 - ✓ Passive smoking(smoker parents)
 - ✓ Pulmonal comorbidity
 - ✓ Airway, ENT, eye surgery
 - ✓ If endotracheal intubation is needed

CONT'D

- **If the benefits factors are okey next slide**
- ENT surgery strongly indicated (adenoids/tonsils infectious focus)
- Expertise of the anesthetic team in handling a child with a common cold
- Parental compliance

CONT'D

- **Pretreatment with salbutamol**
- Anticholinergic premedication
- Mask ventilation
- Humidification of inspired gases
- Avoid desflurane
- Use LMA
- Use propofol
- Consider lidocaine
- Longer than usual stay in the recovery room should be planned

Severe cold

- Runny nose
- Moist cough
- Wheezing
- Purulent congestion
- **Fever**
- **Malaise**
- **Defer for ≥ 2 weeks unless life threatening emergency situation**
- Re-evaluate after two weeks and if benefit weighs the risk proceed

Preanesthetic Evaluation—Laboratory

- At least most newborns must have a CBC count and glucose level known (due to risk of hypoglycemia)
- Neonatal Hgb is primarily fetal hemoglobin (with high O_2 affinity)
- The hemoglobin dissociation curve is shifted to the left
- It releases less oxygen to the tissues than adult hemoglobin
- Newborns have a higher (15-18g/dL) hemoglobin range than older child
- " Rarely, a newborn will have polycythemia (>20 g/dL hemoglobin)

CONT'D

- Determine RBS at a time close to the proposed surgery
- Preterm or SGA newborn are at particular risk for hypoglycemia
- Expect a glucose level between 60 -80 mg/dL full-term newborn
- Expect a glucose level between 50-70mg/dL for a preterm neonate

CONT'D

- Children very susceptible to hypoglycemia and require frequent monitoring and modification of IV fluids are:
 - ✓ Those from a diabetic mothers
 - ✓ Those who haven't been receiving either enteral or parenteral feeds
 - ✓ Those who are very low birth weight
 - ✓ Those who have been septic

CONT'D

- An **innocent** murmurs occurs in **> 30%** of normal children
- An echocardiogram should be obtained if:
 - The child is symptomatic (eg. poor feeding, failure to thrive, or easy fatigability)
 - The murmur is harsh, loud, holosystolic, diastolic, or
 - The murmur radiates widely; or pulses are either bounding (eg, with aortic runoff lesions) or markedly diminished

CONT'D

- For optimal neonatal anesthesia care, always ensure that there is:
 - A thorough preoperative evaluation and preparation
 - An appropriate monitoring
 - The proper size and variety of equipments available
 - Obtain most intense level of support, both in the operating room and in ICU
- Check the airway, IV access, and all monitoring in advance of surgery

CONT'D

- Carefully calculate drug dose and preparation of drugs
- Carefully administer all medications as needed
- Prevention of paradoxical air emboli is critical for pediatrics (no air in syringes as well as in the infusion fluids)
- ETCO₂ monitoring

CONT'D

- Propofol and IAAs can cause hypotension and bradycardia
- Low conc. of potent IAA can be used with opioids to control the hemodynamic responses
- The merits of one anesthetic technique over another is not well defined

CONT'D

- Regional anesthesia can be highly effective in neonates...PSNS?
- Caudal and spinal anesthesia are relatively straightforward
- Where to safely place a lumbar or thoracic epidural block requires considerable skill
- Epidural local anesthetic infusions can result in LAA systemic toxicity

CONT'D

- The more a newborn has received parenteral fluids, the more the chance of electrolyte abnormalities (immature kidney, ongoing losses with replacement mismatch)

CONT'D

- Extremely preterm(<27 week GA at birth) infants, are capable of feeling pain
- Indeed, preterm infants may be even more sensitive to painful stimuli
- Neonates cannot form **explicit memory** for subsequent procedures
- No child should be denied analgesia (anesthesia) due to size or age

PERIOPERATIVE ANXIETY IN CHILDREN

- Children can experience significant anxiety & stress before anesthesia
- There is association between **preoperative anxiety** and adverse postoperative outcomes:
 - ✓ Emergence delirium
 - ✓ An increased analgesic requirement
 - ✓ Negative behavioral changes

CONT'D

- In general, fear in children can be related to a number of factors:
 - ✓ Parental separation
 - ✓ Un unfamiliar and threatening hospital environment
 - ✓ Painful procedures
 - ✓ The operation itself and anesthesia
- Infants up to 9 months of age are less prone to separation anxiety, and will most probably accept parental surrogates

CONT'D

- Separation anxiety is the greatest for children between 1-3 yrs of age
- These children may respond to distraction techniques such as toys and stories
- Parental presence at anesthesia induction has been advocated
- Children between 3 and 6 years of age have concerns about body mutilation
- Preoperative play therapy is especially useful in this age group
- Children between 7 and 12 years of age usually require more explanation

CONT'D

- Videos, leaflets, and interactive computer apps, cartoon movies
- On the day of intervention: use of tablets, arrival to operating room in toy cars, etc., are equal or better than premedication in studies
- Hypnosis, music, and non-aggressive lighting can also be used to provide a calm and soothing environment for the child upon arrival to the operating room

Preanesthetic fasting

- Incidence of aspiration pneumonia in children is 1-2/10,000
- Children with:
 - ✓ Delayed gastric-emptying times (obstruction, trauma, critically ill)
 - ✓ Abdominal pathology like pyloric stenosis, ileus, vomiting,
 - ✓ Electrolyte disorders are at risk of aspiration regardless of fasting duration

CONT'D

- “6-4-2 rule for solid, breast milk and clear fluid ” controversial
- European consensus statement on fasting in children recommends a 1 hr long fasting after clear fluid intake
- Depending on age, regular formula feedings or solid foods are continued until 4–8 hr before surgery(next slide)

CONT'D

- **More specifically,**
- Infants younger than 6 months are fed formula up to 4 h before induction
- From 6–36 months can be given formula or solids up to 6h before induction
- Clear fluids are offered until 2–3 h before induction
- **Must be first on the operation schedule list, why?**

Preanesthetic Plan

- Major factors that should be considered in planning the anesthetic are:
 1. The need to have blood and blood products available before beginning the case
 2. The need for invasive monitoring
 3. The need for additional equipment for the airway or the vascular access
 4. The need to transport the child to and from the operating room
 5. The likelihood of postoperative **ventilation...NICU/PICU**
 6. The plan for postoperative pain relief

CONT'D

- Medical status and the planned surgery will impact this plan
- We have to consult specialities ahead of final plan
- Occasionally, as planning progresses, re-evaluation and further resuscitation may be prudent with the procedure
- After clear anesthetic plan, discuss and take an informed consent from a parent or caregiver

CONT'D

- Informed consent is a process by which the anesthetist explains his/her understanding of the patient's status, planned procedure, the plan for anesthetic mgt, alternatives to the plan, and discussion of risks and benefits
- The goal of informed consent is to help the parent understand what care is being proposed, the risks and benefits involved, and reasonable alternatives
- The basis of true informed consent is the discussion, in terms they understand

Premedication

- Premedication is not commonly used for neonatal anesthesia mgt
- Sedation is not usually appropriate for neonates and sick infants
- Analgesics are rarely indicated before arrival to OR
- IAA induction was associated with bradycardia and hypotension
- Atropine pretreatment was advocated previously
- Now, using opioids as the basis of the anesthetic, with lower doses of IAAs titrated to the effect is advocated

CONT'D

- Midazolam is the most commonly used as a result of its profile of safety and efficacy
- Oral dose= 0.5 mg/kg (max. 15 mg) produces sedation and anxiolysis in 25 min of admi
- IV dose of =0.05-0.1 mg/kg
- Intranasal dose of =0.3 mg/kg
- Rectal dose of =0.5 mg/kg
- Potential limitations is the potential for prolonged effect

CONT'D

- Fentanyl as a lollipop (Actiq 5–15 µg/kg)
- Its levels continue to rise intraoperatively
- This can contribute to postoperative analgesia
- S/E are vomiting, pruritus, and respiratory depression with anesthetics

CONT'D

- Agonists of α_2 adrenergic receptors are also used for premedication
- Clonidine oral (4 $\mu\text{g}/\text{kg}$) or intranasal (4 $\mu\text{g}/\text{kg}$) has long onset
- Its **analgesic and anesthetic**-sparing properties are desired
- **Dexmedetomidine** has a shorter onset and offset time than clonidine
- This makes it an interesting alternative for premedication

CONT'D

- Ketamine is rapidly absorbed after either oral, intranasal, IM, or IV routes
- Oral dose = 5-8 mg/kg, onset time of sedation in 15-20 minutes
- IM dose = 4-5 mg/kg in uncooperative child when the procedure cannot be delayed or rescheduled
- Premedication with ketamine is associated with:
 - ✓ Hypersalivation
 - ✓ Hyperventilation
 - ✓ Hallucinations
 - ✓ An increased incidence of emergence delirium

CONT'D

- Remedy for these ketamine side effects
 - ✓ Atropine pretreatment
 - ✓ low dose propofol pretreatment
 - ✓ prevention of crying before induction(psychological mgt)

Intraoperative Considerations

Monitoring

- Similar monitoring devices with adults with modifications
- Monitoring pediatrics is very challenging in anesthesia practice:
- Unavailability of devices (TEE echo, PA catheterization, and EEG) or
- Other monitors can't provide reliable information for technical reasons

CONT'D

- The goal of monitoring should be to establish ASA standard monitors:
- Pulse oximetry
- Blood pressure
- ECG
- Add invasive monitoring, as appropriate
- But, get the appropriate sizes for each monitoring devices

CONT'D

- ETCO₂ or Transcutaneous CO₂ may be monitored for diagnosing changes over time, bronchospasm, a kinked endotracheal tube, endobronchial intubation
- **Hypocarbica** should be avoided due to the risk of cerebral vasoconstriction

CONT'D

- Pulse oximetry is mandatory to detect hypoxia, and helps avoid hyperoxia
- The optimal oxygen saturation for neonates is controversial retinopathy Vs neurologic morbidity
- Scholars found that, aiming for low (85%-89%) saturations in **extremely preterm** infants may increase the risk of other neurologic morbidity

CONT'D

- During stable situations it is reasonable to aim **SPo2 of 93-95%** in preterm infants
- When managing such a delicate balance we have to be very vigilant and rapidly respond to changes in oxygen saturation

CONT'D

- **Precordial stethoscope** have to be applied always
- It provides inexpensive means of monitoring HR, heart rhythm, quality of heart sounds, breath sounds, and airway patency
- **It is a hallmark of the pediatric anesthesia practice**
- Esophageal stethoscope is more secure and less susceptible to external noise, it also provides core temperature measurement(unlike precordial)

CONT'D

- Clinical observation of the patient's:
 - ✓ Color
 - ✓ Capillary refill
 - ✓ Warmth of skin
 - ✓ Muscle tone
 - ✓ Fullness of fontanelle
 - ✓ Chest expansion are useful monitors

CONT'D

- Large dependence on electronic monitors during procedure
- Probes on either leg may give lower values than a probe on the right hand
- Due to preductal and postductal saturations differences

CONT'D

- Neonatal Hgb in the first 2weeks of life, is a fetal hemoglobin
- The pulse oximeter does not compensate for the leftward shift
- Pulse oximeter values read about 2% higher than arterial blood saturations

CONT'D

- **Capnography waveforms** can give significant information about:
 - ✓ Changes in ventilation
 - ✓ Obstruction
 - ✓ Rebreathing
 - ✓ Adequacy of ventilation
 - ✓ Confirmation of endotracheal tube placement
 - ✓ Early warning of malignant hyperthermia

CONT'D

- Airway pressure measurements are useful to adopt adult anesthesia ventilators for use in neonatal and pediatric patients
- PAW is used as a guide for setting tidal volume

Temperature

- Must be closely monitored in pediatric patients:
 - ✓ Higher risk for malignant hyperthermia
 - ✓ Potential for both iatrogenic hypothermia and hyperthermia

CONT'D

- Hypothermia can be prevented by maintaining:
 - ✓ A warm operating room environment (26°C or higher)
 - ✓ Warming and humidifying inspired gases
 - ✓ Using a warming blanket and warming lights
 - ✓ warming all intravenous fluids

CONT'D

- The room temperature required for a neutral thermal environment is highest in premature newborns
- Note that care must be taken to prevent unintentional skin burns and iatrogenic hyperthermia from overzealous warming efforts

Glucose

- Premature or SGA neonates who have received **hyperalimentation**, or **who are from diabetic mothers are** be prone to hypoglycemia
- These infants should have frequent serum glucose determinations:
- Levels **< 30 mg/dL in the neonate** and
- Levels **< 40 mg/dL in older children** indicate hypoglycemia

Anesthetic Systems

- It provides final conduit for delivery of anesthetic gases to a patient
- Breathing circuits link a patient to an anesthesia machine
- Many modifications in circuit design have been developed, each with varying degrees of efficiency, convenience, and complexity

Open-Drop Anesthesia

- A highly volatile anesthetic(ether or halothane) is dripped onto a gauze covered mask applied to the patient's face
- Air passes through the gauze on inhalation, vaporizes the liquid agent, and carries high concentrations of anesthetic to the patient

CONT'D

- The vaporization lowers mask temperature, resulting in moisture condensation and a drop in anesthetic vapor pressure
- This system is not used in modern medicine
- Modern draw-over vaporizers are its derivatives

Draw-Over Anesthesia

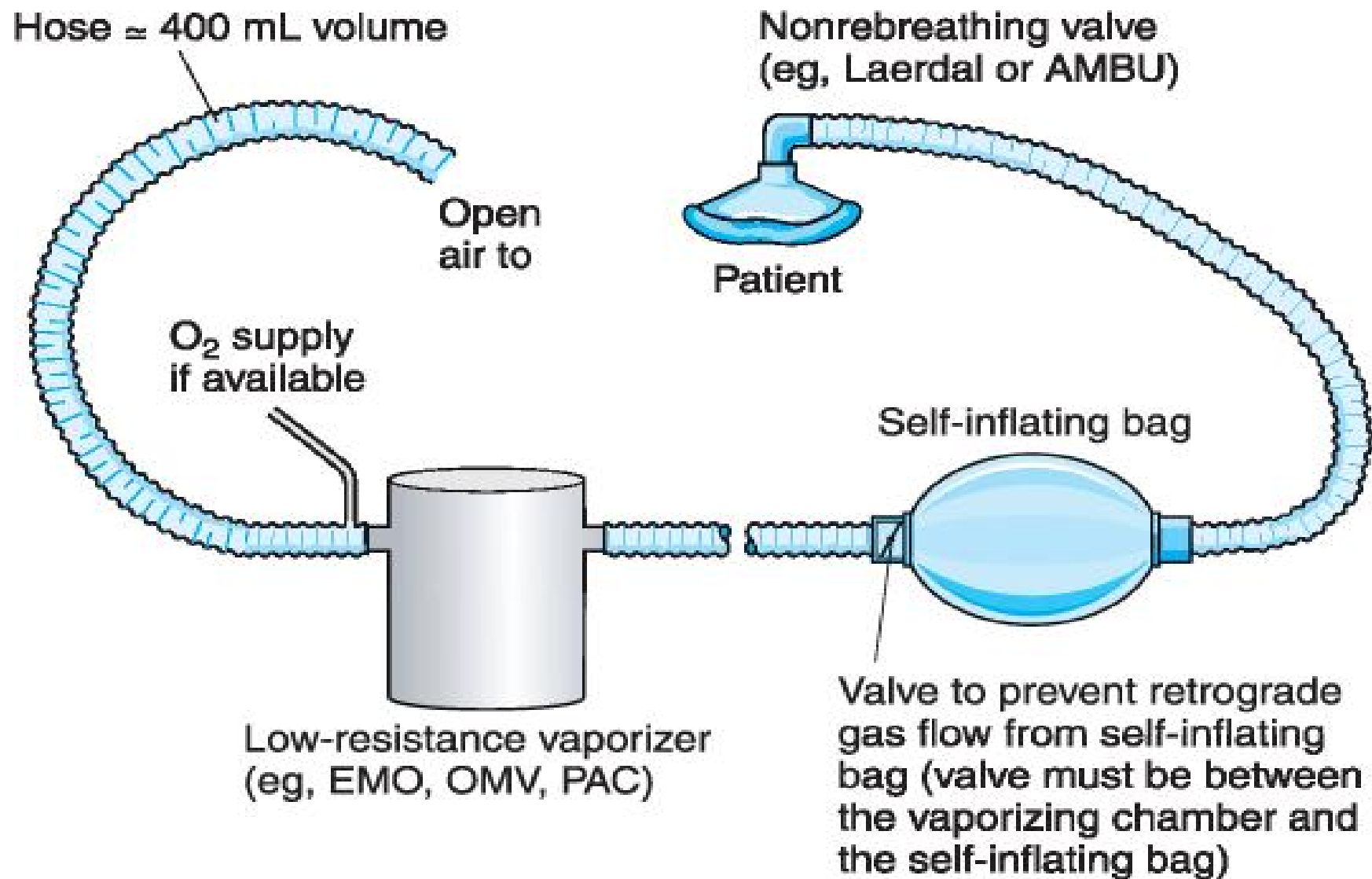
- It has nonrebreathing circuits that use ambient air as the carrier gas
- Supplemental oxygen can be used if available
- Simple, predictable and controllable inspired vapor and O₂ concentration
- The devices can be fitted with connections and equipment that allow IPPV and passive scavenging, as well as CPAP and PEEP

CONT'D

- Air is drawn through a low-resistance vaporizer at inhalation
- Patients spontaneously breathing room air and IAAs (except N₂O) often manifest a saturation of (SpO₂) < 90%
- This can be treated with IPPV, supplemental oxygen, or both
- It has an open-ended reservoir tube of about 400 mL, attached to a T-piece at the upstream side of the vaporizer (figure next slide)

Good properties of draw over devices

- ✓ Portable
 - ✓ Robust
 - ✓ Low resistance to gas flow
 - ✓ Usable with any agent
 - ✓ Controllable vapor output



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Disadvantages

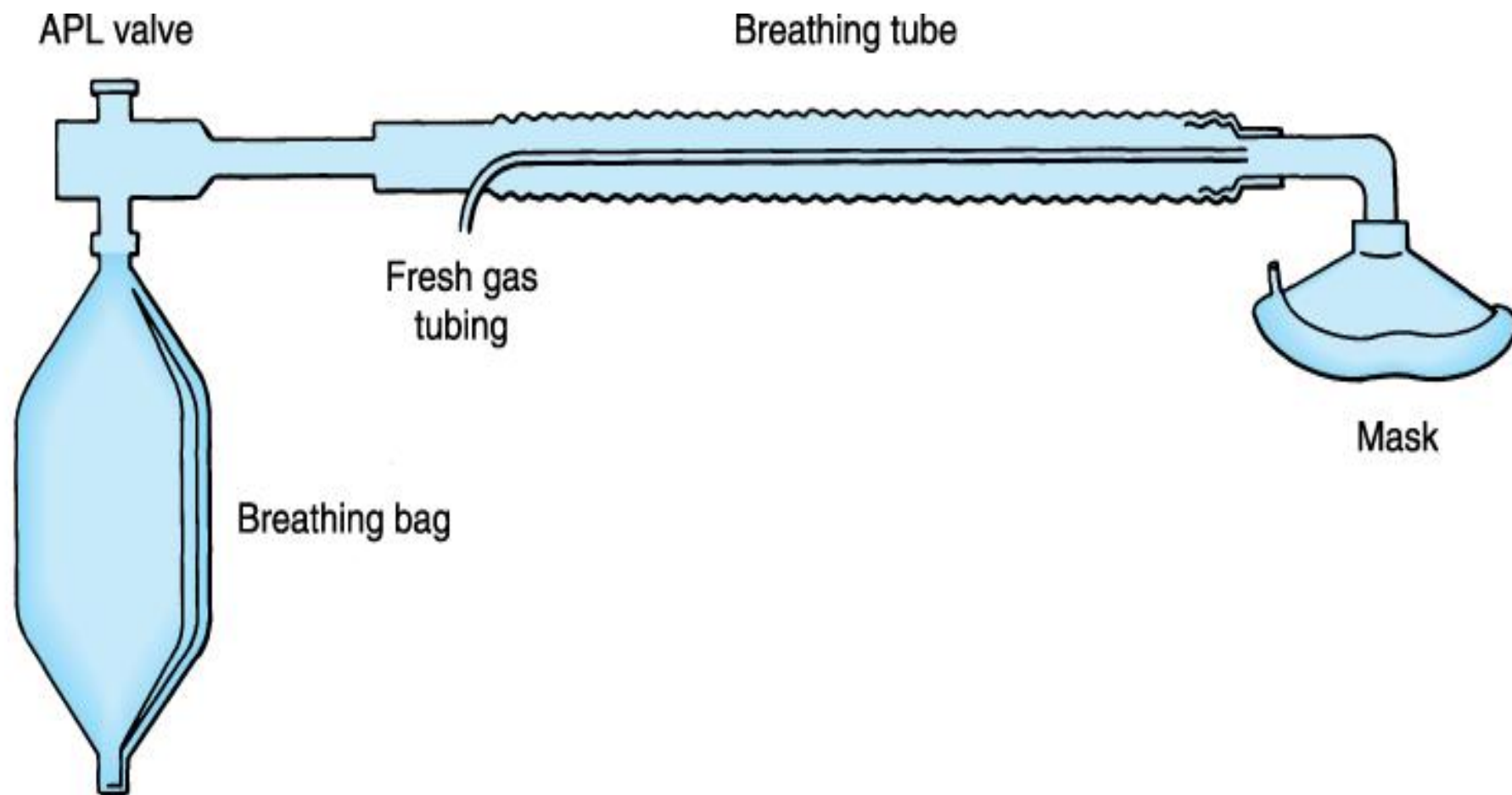
- Absence of a reservoir bag (depth of TV is not well known during SV)
- The nonrebreathing valve, PEEP valve, and circuit filter close to the patient's head makes the technique hard to use for head and neck surgery and pediatric cases
- If the head is draped, the nonrebreathing valve is often covered
- OR pollution with large volumes of waste gas

Mapleson Circuits

- This system solve some of the problems of former devices
- It incorporates breathing tubes, fresh gas inlets, APL valves, and reservoir bags) into the breathing circuit
- These components relative location determines circuit performance and is the basis of the Mapleson classification
- Its efficiency depends of high fresh gas flow rates

Bain circuit

- Modification of the Mapleson D system that incorporates the fresh gas inlet tubing inside the breathing tube (Figure below)
- The modification decreases the circuit's:
 - ✓ Bulk and retains heat and humidity better a result of partial warming of the inspiratory gas by countercurrent exchange with the warmer expired gases



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CONT'D

- Disadvantage of this coaxial circuit is:
 - ✓ Possibility of kinking
 - ✓ Disconnection of the fresh gas inlet tubing
 - ✓ These mishaps could result in significant rebreathing of exhaled gas

The Circle System

- In an attempt to avoid problems of maplsons circuite
- This circle system adds more components to the breathing system
 - ✓Carbon Dioxide Absorbent
 - ✓Unidirectional Valves

Disadvantages of the Circle System

- Greater size and less portability
- Increased complexity
- Higher risk of disconnection or malfunction
- Increased resistance
- Difficulty of predicting inspired gas conc. during low fresh gas flows

ANESTHESIA INDUCTION

CONT'D

- No one best method of induction and maintenance of anesthesia for all
- Method of inducing anesthesia is determined by the:
 - ✓ Current medical status of the patient
 - ✓ Surgical condition or surgical procedure
 - ✓ Presence of ongoing fluid or blood losses
 - ✓ Gestational age of the child

CONT'D

- ✓ Level of anxiety of the child
- ✓ Child's ability to cooperate and communicate
- ✓ Presence or absence of a full stomach
- ✓ Recent fasting, and
- ✓ Experience of the anesthesia professional

CONT'D

- Parents and OR staff should manage combatant child perioperatively
- Most neonates who come to OR have an IV secured
- It is must to secure IV line before induction for a neonate who is:
 - Preterm
 - medically unstable
 - has a full stomach
 - Has a potentially difficult airway
 - Has ongoing fluid losses

CONT'D

- For older children with no IV access inhalational induction
 - High conc. of sevoflurane inhalational induction or
 - Ketamine (5-7 mg/kg IM) can be helpful
- Recent studies recommend that the IV induction is better than inhalational one with respect to PRAE
- Hence, When deciding on the induction technique, care should be taken to weigh all the relevant factors

CONT'D

- Prior application of EMLA for IV cannulation
- Intraosseous administration may also be a possibility
- But, some may be very anxious at the needle sight
- Awake or sedated-awake intubation with topical anesthesia should be considered for emergency surgery
- For critically ill or a potential difficult airway is present

INTRAVENOUS INDUCTION

Rapid sequence induction(RSI)

- Is used to prevent pulmonary aspiration of gastric content
- The “classical form” RSI use to pediatrics may not always be the correct choice due to:
 - Anatomical and physiological differences
 - Adequate preoxygenation cannot be achieved in **children? (fast desaturation, crying)**
 - Preoxygenation may not be effective like adults due to the low FRC(even cooperative)

CONT'D

- Brief periods of apnea in the absence of PPV can lead to profound hypoxemia and related bradycardia
- Application of cricoid pressure may distort the airway in young children making glottic visualization more difficult
- These factors may lead to a higher incidence of forced mask ventilation and unsuccessful intubation attempts

CONT'D

- Remedy:
- Balance aspiration risk with more prevalent risk of hypoxemia by using **controlled forms of the RSI technique**
- Also reduce the risk of regurgitation and vomiting from direct laryngoscopy by deep anesthesia and complete muscle paralysis

CONT'D

- With an IV line insitu:
 - Rapid induction of adequate hypnosis infact with analgesia
 - Profound muscle paralysis using a NDMR
 - Gentle mask ventilation with a maximum PAW of 12 cm H2O until endotracheal intubation can be performed

CONT'D

- The adoption of a “controlled” approach may reduce:
 - The potentially significant risk of hypoxemia
 - While providing rapid intubating conditions

CONT'D

- The induction sequence is the same as used in adults
 - ✓ Adequate analgesic and anti-emetics
 - ✓ A rapid-acting barbiturate (Thiop=3 mg/kg IV neonates, 5–6 mg/kg IV in infants)
 - ✓ Propofol (2–3 mg/kg)
 - ✓ A nondepolarizing muscle relaxant or succinylcholine

CONT'D

- Atropine should be given IV prior to succinylcholine
- With the use of propofol as an induction agent:
 - Less hypertension during intubation
 - A faster and smooth onset and awakening
 - Less postoperative nausea and vomiting

CONT'D

- The advantages of an intravenous technique includes:
 - Familiarity with the induction agents
 - Availability of intravenous access if emergency drugs needed
 - Rapidity of induction in the child at risk for aspiration

Inhalational Induction

CONT'D

- They (pediatrics) mostly arrive in the OR without an IV line
- Modern potent IAAs can render them unconscious shortly
- It is easier in children who are sedated prior to entering the OR and who are sleepy enough to be anesthetized

CONT'D

- Alternatives include:
 - Insufflation of the anesthetic gases over the face
 - Using a substituted clear face mask
 - Placing a food flavor on the inside mask (eg, oil of orange)
 - Allowing a child to sit during the early stages of induction

CONT'D

- Many reasons for faster uptake of anesthetics in infants:
 - The 5:1 alveolar ventilation to FRC ratio (1.5:1 in adult)
 - More of CO goes to the VRG of organs(heart and brain)
 - The neonate has a greater CO per kg of body mass
 - They have a lower blood gas partition coefficient for IAAs
 - Use of nonrebreathing systems (Bain or Mapleson "D" circuit)

AIRWAY MANAGEMENT AND VENTILATION

AIRWAY MANAGEMENT

- Preoperative airway evaluation should be based on both the child's medical record and clinical evaluation
- On hx check for birth complications, OSA, head and neck trauma, previous surgery, and related airway mgt
- On P/E-check for facial dysmorphias, signs of stridor, dysphonia, swallowing disorders, difficulty in breathing, and hoarseness

CONT'D

- There are many difficult AW predictors in children, their applicability, sensitivity, and specificity vary greatly in settings
 - Mandibular protrusion
 - Mallampati's classification
 - Movement of atlantooccipital joint
 - Reduced mandibular space
 - Increased tongue thickness

CONT'D

- Other reported risk factors are:
 - ✓ Age < 1 year
 - ✓ ASA II-IV status
 - ✓ Obesity
 - ✓ Maxillofacial and cardiac surgery

CONT'D

- Anesthetized children are particularly prone to UAWO
- These can be easily relieved by a combination of:
 - Moderate head tilt, chin lift, jaw thrust
 - Application of continuous positive airway pressure
 - Lateral decubitus position (recovery position)

A facemask airway

- This technique can be successfully be used for a brief time
 - The tolerances of mask fit
 - Adequate airway pressure
 - Avoidance of gastric distention are small
 - Making this a poor choice for any but the briefest operations

CONT'D

- **Controlled ventilation** is more commonly used than spontaneous ventilation for surgical procedures
- An artificial airway is necessary
- Plus, both **OPA and NPA** devices can be used during SV or PP mask ventilation to relieve AWO from posterior displacement of the tongue in anesthetized children

CONT'D

- PPV during mask induction and prior to intubation may causes gastric distention, resulting in impairment of lung expansion
- Suctioning with NG or OG tube will decompress the stomach, but it must be done carefully to avoid mucous membrane injury

Laryngeal mask airway(LMA)

- It is used to provide ventilation for newborn as small as 1 kg
- While preparing to use other intubating techniques
- LMA can be used as a guide for blind intubations in newborns with the use of a stylet tube

CONT'D

- Fiberoptic laryngoscopy, is routinely used in older children and adults, can also be used in the newborn
- Fiberscopes are currently available that accept endotracheal tubes as small as 2.5 -mm ID
- These scopes do not currently have ability to change direction and are useful more for confirmation of tube placement

CONT'D

- Those changing direction accept at least 3.5-mm ID tube size
- Fiberscope insertion can be done directly or through an LMA
- An LMA guide has been useful in directly intubating newborns that could not be visualized by routine approaches

CONT'D

- The two commonly used LMA are
- The classical LMA and the LMA ProSeal
- Evidence shows LMA use in children is associated with a decreased incidence of PRAE when compared to ETT insertion

CONT'D

- LMA size fit is best estimated on the per kg basis
- It should be used for shorter procedures
- Caution should be applied while using LMA
- It is not appropriate for oral procedures
- Its position should be well fixed and continuously handled
- Associated with high incidence of laryngospasm in children

	Premature	Neonate	Infant	Toddler	Small Child	Large Child
Age	0–1 month	0–1 month	1–12 months	1–3 years	3–8 years	8–12 years
Weight (kg)	0.5–3	3–5	4–10	8–16	14–30	25–50
ETT (mm ID)	2.5–3	3–3.5	3.5–4	4–4.5	4.5–5.5	5.5–6 cuffed)
ET depth (cm at lips)	6–9	9–10	10–12	12–14	14–16	16–18
Suction catheter (F)	6	6	8	8	10	12
Laryngoscope blade	00	0	1	1.5	2	3
Mask size	00	0	0	1	2	3
Oral airway	00	00	0 (40 mm)	1 (50 mm)	2 (70 mm)	3 (80 mm)
LMA	—	1	1	2	2.5	3

Tracheal Intubation

- Tracheal intubation is done under direct laryngoscopy
- Awake intubation may be safe without a danger of AW loss
- But, it can be a traumatic experience due to:
 - An accompanied pain
 - Breath holding
 - Desaturation
 - Tissue trauma

CONT'D

- The likely desaturation can be prevented by using an oxyscope
- Reserved for patients of severe hemodynamic compromise:
 - An extraordinarily distended and tense abdomen
 - Presumed difficult airway (e.g newborn with micrognathia)
- Addition of sedation with opioid or topical application of LA can help decrease procedural trauma

CONT'D

- It has also been suggested that an awake intubation may be best for the anesthetist with less experience in intubating newborns
- It may be better to have a more experienced clinician, if available, attend to the airway in that situation

CONT'D

- Most newborns are intubated after a RSI (controlled)
- Preoxygenation is useful in adding additional safety
- The concern is about a period of hyperoxia in a preterm
- No adequate evidence for preoxygenation to cause retinopathy
- For **suspected difficult intubation**, induce anesthesia, adequacy of mask ventilation, and then give muscle relaxant

CONT'D

- Positioning for intubation depends on neonatal AW differences
- The large Occiput, already has a **flexed neck**
- Additional extension may be done by a **shoulder roll**
- A Miller blade no. 0 for the preterm and no. 1 for full-term newborn is usually used
- ETT ID no 2.5- 3 for premature, 3-3.5 newborn and 3.5-4 infants

CONT'D

- The ID of the ETT estimated by a formula based on age:
- $\text{Age}/4+4$
- For example, a 4-year-old child would be predicted as
- $4/4+4=5$ ETT ID
- Prepare ETT 0.5 mm larger and smaller than predicted
- Uncuffed ETT are used for children up to 8–10 years (next slide)

	Premature	Neonate	Infant	Toddler	Small Child	Large Child
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Oral airway	00	00	0 (40 mm)	1 (50 mm)	2 (70 mm)	3 (80 mm)
LMA	—	1	1	2	2.5	3

CONT'D

- To decrease POP croup and provide leak to prevent barotrauma
- No leak indicates a need for replacement to prevent POP edema
- Excessive leak may avoids adequate ventilation and contaminate the operating room with anesthetic gases
- For younger patients **at risk of aspiration**, use a down-sized cuffed tube with the cuff minimally inflated

CONT'D

- There is also a formula to estimate endotracheal length:
- $\text{Age}/2 + 12 = \text{ETT depth at the lips}$
- for 1 year child this will be at 12.5 cm depth at the lips
- The calculations are estimates and prematures, neonates and infants are exceptions (next slide)

	Premature	Neonate	Infant	Toddler	Small Child	Large Child
Age	0–1 month	0–1 month	1–12 months	1–3 years	3–8 years	8–12 years
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LMA	—	1	1	2	2.5	3

CONT'D

- Every patient's anatomy is different
- But, for best visualization advance the laryngoscope in a direction parallel to the handle
- If the glottis is not easily seen:
 - Apply cricoid pressure by a little finger holding the handle
 - By well trained assistant

CONT'D

- Once the ETT is inserted check for:
 - The presence of a positive capnograph tracing
 - A bilateral expansion of the thorax
 - Bilateral breath sounds are used to see proper placement
 - Listen for an air leak at an airway pressure of about 10-20 cm H₂O
 - Clinical judgement and auscultation may also confirm it
- Some may prefer to advance the ETT past the carina and then withdraw until bilateral breath sounds are heard

CONT'D

- A cuffed ETT has been commonly used in neonates, infants, and young children
- Its preferred in emergency and in a risk for aspiration group
- Uncuffed ETT had higher incidence of laryngospasm
- For cuffed ETT to cause higher subglottic edema there is no adequate data

CONT'D

- A cuffed tube with no leak may also allow a more accurate estimate of ETCO_2 concentrations
- Avoids pollution of the operating room
- Using cuffed ETT prevents the need for frequent changing of uncuffed ETT due to significant leak linked with it
- Repeated laryngoscopy is avoided since with cuffed ETT

VENTILATION

- Once the AW is secured, ventilation may be controlled during neonatal surgical procedures with hand ventilation or MV
- After establishing a baseline acceptable ventilation
 - Then continuously monitor the peak airway pressures
 - Monitor for chest expansion
 - Monitor for return volume
 - Monitor for pulse oximetry
 - Monitor for capnograph tracings for changes

CONT'D

- For significant changes in compliance and ventilation:
 - Underlying pulmonary disease
 - A shift of ETT
 - Surgical manipulation can be responsible
- Initial tidal volumes of 10 ml/kg
- RR=20 -25 breaths per minute are a reasonable

CONT'D

- Recent international guidelines for mgt of unanticipated DAW in pediatrics focuses on airway mgt of children between 1-8yrs
- Three scenarios are identified:
 - a) Difficult mask ventilation
 - b) Difficult tracheal intubation
 - c) Cannot intubate and cannot ventilate in paralyzed anesthetized children

CONT'D

- The suction catheter is the smallest for a neonates
- The face mask is also smallest for neonates
- The laryngoscope is also very smallest for newborns
- These all differences should be considered when preparing to anesthetize very young children

CONT'D

- Then expected the positive airway pressures=20 cm H₂O
- Expect a level of PEEP =3- 5 cm H₂O (prevents atelectasis)
- For a patient with pulmonary disease, set higher volumes
- For a patient previously ventilated in NICU with a ventilator modality not supported by the anesthesia ventilator, prepare and have those modalities in the OR

Maintenance

- After airway is managed effectively, following induction
- Control ventilation during anesthesia of neonates and infants
- For SV, low resistance of a circle system can be a significant obstacle for a sick neonate to overcome

CONT'D

- The resistance is from unidirectional valves, breathing tubes, and absorbers
- Some may prefer the Mapleson D circuit or the Bain system for patients weighing <10 kg (low resistance and light weight)
- **Circle system** can be safely used in patients of all ages if **ventilation is controlled**

CONT'D

- Monitor AW pressure to r/o kinking or endobrocheal intubation
- Many previous ventilators cannot provide the low TV and rapid rates required by neonates and infants
- Unintentional large TV delivery may cause high PAW pressure and barotrauma
- The **pressure-limited mode** should be used for neonates, infants, and toddlers

CONT'D

- Deliver small TV manually by using 1L child bag
- For a children < 10 kg, adequate TV is achieved with peak inspiratory pressures of 15–18 cm H₂O
- For **larger children use VC** ventilation (TV=8–10 mL/kg)
- Equipment dead space, highly determines rebreathing in children
- An appropriately sized, pediatric HME should be employed

CONT'D

- Maintain anesthesia with the same agents as in adults
- Switch to isoflurane or halothane after **sevoflurane induction**
- This reduces the likelihood of postoperative delirium or agitation on emergence
- If sevoflurane is used for maintenance, an opioid (eg, fentanyl 1–1.5 $\mu\text{g}/\text{kg}$) 15–20 min before procedure ends
- Reduces incidence of emergence delirium and agitation

CONT'D

- The MAC of IAAs is higher in children than in adults
- Neonates may be particularly susceptible to cardiodepressant effects of general anesthetics
- NDMR are often required for optimal surgical condition
- Particularly for neonates and sick infants who may not tolerate higher doses of volatile agents

Emergence and Recovery

- Hypothermia is a common perioperative problem in children
- Inability to regulate body temperature under GA, cold large OR, and continued heat loss are major reasons for hypothermia
- Minor hypothermia (34 to 36°C) has not been found to influence the recovery period

CONT'D

- Assure normothermia before discharge, as more significant hypothermia can result in:
 - Increased oxygen consumption
 - Cardiovascular manifestations
 - Prolonged metabolism and excretion of drugs
 - Delayed wound healing

CONT'D

- Emergence agitation is presence of thrashing, crying, screaming, and disorientation during emergence from anesthesia in any age
- It is more common in pediatrics, particularly who had strabismus surgery or tonsillectomy or adenoidectomy
- Clearly lower incidence after propofol-based anesthesia
- Treatment with fentanyl, dexmedetomidine, midazolam
- Symptoms resolve 30-45minutes after surgery

Extubation Criteria

- Follow extubation criteria established for adults
- The child's protective airway reflexes are returned
- Make sure the child's breathing pattern is regular
- Make sure the child's breathing pattern is adequate
- Make sure the child's breathing pattern is synchronized
- Make sure the child's breathing is within normal range

CONT'D

- The child should maintain its saturation spontaneously
- The child should also maintain SpO₂ of at least 95% on O₂
- The child's airway is clear
- The child is adequately reversed from NMBDs
- Get ready for unanticipated difficult airway (pedi airway)
- Make sure the child cries before handover to the PACU

Postoperative nausea and vomiting (PONV)

- PONV occur frequently after eye or ear surgery
- Some studies quote an overall incidence of 20 to 30%
- while others have found that it's about 39 to 73%
- The mechanisms for PONV are central causes (opiates), gastrointestinal malfunction (ileus or gastric distention), the surgical procedure itself (eye and ear), and postoperative pain

CONT'D

- Control of nausea and vomiting should be part of the selection of agents/techniques used for anesthesia
- Use of propofol based anesthesia, anti-emetics and adequate pain management are recommended
- Make sure the airway is clear and breathing is adequate before handover to PACU

CONT'D

- Pretreatment with **ondansetron=0.15** mg/kg or plasil=0.15 mg/kg successfully reduces PONV for patients at higher risk
- Some children may experience nightmares or have behavioral problems after a surgical procedures
- Many institutions have found that allowing parents to soothe the child during recovery is beneficial

CONT'D

- The other most common postoperative complications in pediatrics are laryngospasm and postintubation croup
- Like in adults post operative pain should be well managed

Laryngospasm

- It is a forceful, involuntary spasm of the laryngeal muscles caused by superior laryngeal nerve stimulation
- It may occur at induction, emergence, or any time in between without an ETT in place
- It is more common in young pediatric patients than in adults, being highest in infants 1–3 months old
- It can be avoided by either or deep extubation during recovery

CONT'D

- Extubation between between these extremes, however, is generally recognized as hazardous
- A recent URI or being secondary smoker predisposes patients to laryngospasm on emergence

CONT'D

- **Remedy:**
- Gentle PPV, forward jaw thrust, lidocaine=1–1.5 mg/kg IV, or paralysis with succ=0.5–1 mg/kg IV or rocuronium= 0.4 mg/kg and controlled ventilation
- Succ=4–6 mg/kgIM is acceptable alternative if no IV access and in whom more conservative measures are failed

CONT'D

- It is usually an immediate postop event but may occur in the recovery room as well
- Hence, recovering pediatrics should be placed in lateral position so that oral secretions pool and drain away from the vocal cords
- As the child begins to regain consciousness, it is comforting to have the parents at the bedside

Postintubation Croup

- Croup is due to glottic or tracheal edema
- The narrowest part of pediatric airway is cricoid cartilage
- It is less common with uncuffed or smaller ETT tubes

CONT'D

- Postintubation croup is associated with:
 - Early childhood (age 1–4 years)
 - Repeated intubation attempts
 - Large endotracheal tubes
 - Prolonged surgery, head and neck procedures
 - Excessive movement of the tube

CONT'D

- Remedy:
- Dexamethasone=0.25–0.5 mg/kgIV prevents edema formation
- Inhalation of nebulized epinephrine=0.25–0.5 mL of a 2.25% solution in 2.5 mL normal saline (is effective Tx)
- Postintubation croup almost always appears within 3 hrs after extubation

Postoperative Pain Management

- Pain in pediatrics has received high attention in recent years
- Particularly the use of regional anesthetic techniques
- Commonly used parenteral opioids include:
 - Fentanyl=1–2 µg/kg
 - Morphine =0.05–0.1 mg/kg
 - Hydromorphone= 0.015 mg/kg
 - Meperidine=0.5 mg/kg

CONT'D

- Ketorolac=0.5–0.75 mg/kg lowers opioid requirements
- Rectal acetaminophen=40 mg/kg may also be helpful
- Patient-controlled analgesia(PCA) depends on child maturity
- Morphine 20 µg/kg/hydromorphone 5 µg/kg in 10-min lockout
- Infusion of morphine= 0–12 g/kg/h or hydromorphone=0-3 g/kg/h at the expense of respiratory depression

CONT'D

- Like adults, epidural infusions for postoperative analgesia usually consist of a local anesthetic plus an opioid
- Bupivacaine 0.1–0.125% is used with fentanyl 2–2.5 g/mL
- Recommended infusion rates depend on the:
 - Patient size
 - Final drug concentration
 - Location of the epidural catheter,
 - It usually ranges from 0.1 to 0.4 mL/kg/h.

Thank you

Any question?

perioperative problems in anesthesia for pediatrics

CONT'D

- Common peri-operative pediatrics complications include:
 - Difficult Air way, URI, POA, OSA, Asthma, common cold
 - hypoxemia
 - bradycardia
 - hypoglycemia
 - hypothermia
 - hypo/hypervolemia
 - Electrolyte imbalances

Difficult Air way

- Difficult mask ventilation (MV) – during routine induction of anaesthesia in a child aged 1 to 8 years
- Difficult MV → Give 100% oxygen → Call for help

CONT'D

- **Step A =Optimise head position**
- Consider:
 - Adjusting chin lift/jaw thrust
 - Inserting shoulder roll if <2 years
 - Neutral head position if >2 years
 - Adjusting cricoid pressure if used
 - Ventilating using two person bag mask technique

CONT'D

- **Check equipment**
- Consider changing:
 - Circuit
 - Mask
 - Connectors
 - If equipment failure is suspected, change to self-inflating bag and
 - Isolate from anaesthetic machine promptly

CONT'D

- **Depth of anaesthesia**
 - Consider deepening anaesthesia
 - Use CPAP

CONT'D

- **Step B Insert OPA- call for help if not arrived**
- Assess for cause of difficult mask ventilation
 - Light anaesthesia
 - Laryngospasm
 - Gastric distension – pass OG/NG tube

CONT'D

- **Maintain anaesthesia/CPAP**
- Deepen anaesthesia (Propofol first line)
 - If relaxant given – intubate
 - If intubation not successful, go to unanticipated DAW algorithm

CONT'D

- **Step C Second-line: Insert SAD (e.g. LMA)**
 - Insert SAD (e.g. LMA) – not > 3 attempts
 - Consider nasopharyngeal airway
 - Release cricoid pressure
 - Good airway →yes→continoue

CONT'D

- Good airway →no→SpO $>80\%$
- Consider:
 - SAD (e.g. LMA) malposition/blockage
 - Equipment malfunction
 - Bronchospasm
 - Pneumothorax
- Wake up patient

CONT'D

- Good airway →no→SpO $<80\%$
- Attempt intubation
- Consider paralysis →Succed→Proceed
- Good airway →no→SpO $<80\%$
- Attempt intubation
- Consider paralysis →Fail→Go to CICO scenario

Unanticipated DAW algorizm

- Difficult direct laryngoscopy → Give 100% oxygen and maintain anaesthesia → Call for help
- **Step A Initial tracheal intubation plan when mask ventilation is satisfactory**
- Ensure: Oxygenation, anaesthesia, CPAP, management of gastric distension with OG/NG tube

CONT'D

- Direct laryngoscopy – not > 4 attempts
- Check for:
 - Neck flexion and head extension
 - Laryngoscopy technique
 - External laryngeal manipulation – remove or adjust
 - Vocal cords open and immobile (adequate paralysis)
- If poor view → bougie, straight blade laryngoscope and/or smaller ETT

CONT'D

- Succeed→Tracheal intubation →
 - Verify ETT position
 - Capnography
 - Visual if possible
 - Auscultation
- If ETT too small consider using throat pack and tie to ETT
- If in doubt, take ETT out

CONT'D

- Failed intubation with good oxygenation→
- **Step B Secondary tracheal intubation plan →Call for help again if not arrived**
 - Insert SAD (e.g. LMA) – not > 3 attempts
 - Oxygenate and ventilate
 - Consider increasing size of SAD (e.g. LMA) once if ventilation inadequate

CONT'D

- Succeeded→
- Consider modifying anaesthesia and surgery plan
- Assess safety of proceeding with surgery using a SAD (e.g. LMA)→Unsafe→Postpone surgery→Wake up patient
- Safe→Proceed with surgery
- Safe→Consider 1 attempt via SAD →Verify intubation, leave SAD in place and proceed with surgery
- Failed intubation via SAD→Postpone surgery Wake up patient

CONT'D

- Failed oxygenation e.g. SpO $< 90\%$ with FiO 1.0
 - Convert to face mask
 - Optimise head position
 - Oxygenate and ventilate
 - Ventilate using two person bag mask technique, CPAP and OPA/NPA
 - Manage gastric distension with OG/NG tube
 - Reverse non-depolarising relaxant

CONT'D

- Succeeded→Postpone surgery Wake up patient
- Failed ventilation and oxygenation→Go to CICO scenario

CICV scenario

- Failed intubation, inadequate ventilation → Give 100% oxygen → Call for help
- **Step A Continue to attempt oxygenation and ventilation**
 - FiO 1.0
 - Optimise head position and chin lift/jaw thrust
 - Insert oropharyngeal airway or SAD (e.g. LMA)
 - Ventilate using two person bag mask technique
 - Manage gastric distension with an OG/NG tube

CONT'D

- **Step B Attempt wake up if maintaining SpO $>80\%$**
 - If rocuronium or vecuronium used, consider sugammadex (16mg/kg) for full reversal
 - Prepare for rescue techniques in case child deteriorates

CONT'D

- **Step C Airway rescue techniques for CICV (SpO $<80\%$ and falling) and/or heart rate decreasing → call for help again if not arrived**
- Call for specialist ENT assistance → ENT available
- Consider:
 - Surgical tracheostomy
 - Rigid bronchoscopy + ventilate / jet ventilation (pressure limited)

CONT'D

- Call for specialist ENT assistance→ENT not available
- Percutaneous cannula
- cricothyroidotomy / transtracheal jet ventilation (pressure limited)→succeeded
 - Continue jet ventilation set to lowest
 - delivery pressure until wake up or
 - definitive airway established

CONT'D

- Percutaneous cannula
- cricothyroidotomy / transtracheal jet ventilation (pressure limited)→Failed
 - Perform surgical cricothyroidotomy / transtracheal and insertion of ETT / tracheostomy tube
 - Consider passive O₂ insufflation while preparing

Cannula cricothyroidotomy

- Extend the neck (shoulder roll)
- Stabilise larynx with non-dominant hand
- Access the cricothyroidotomy membrane with 14/16G cannula
- Aim in a caudad direction
- Confirm position by air aspiration using a syringe with saline

CONT'D

- Connect to either:
- APL device, set to lowest delivery pressure or
- 4Bar O₂ source with a flowmeter and Y connector
- Cautiously increase inflation pressure/flow rate to achieve adequate chest expansion Wait for full expiration before next inflation
- Maintain upper airway patency to aid expiration

Upper Respiratory Infection (URI)

- History of current URI increases the risk to develop:
 - Laryngospasm (bronchospasm)
 - Oxygen desaturation
 - Postextubation croup
 - Postoperative atelectasis

CONT'D

- A child with asthma, bronchopulmonary dysphasia, under 1 year of age, sickle cell disease, who live with smoking parents, and who undergo bronchoscopy are at a higher risk of developing perioperative morbidity if suffering from a URI
- These patients should be carefully assessed and strong consideration should be given to postponing elective surgery

CONT'D

- Bronchial hyperreactivity may exist for 6-7 weeks after such an infection
- Take the final decision with risk-to-benefit ratio
- The anesthesia technique may impact the number of airway issues encountered

CONT'D

- Mask anesthesia has been linked with lower rate of perioperative complications than ETT
- The LMA use is linked with the same number of AW complications as the endotracheal tube in this population

Postoperative Apnea (POA)

- Apnea and bradycardia are major complications during and after surgery in neonates
- The infants at highest risk are those:
 - Born prematurely
 - With multiple congenital anomalies
 - With a history of apnea and bradycardia
 - With chronic lung disease

CONT'D

- It has a multifactorial etiology
 - ↓sed ventilatory control and ↓sed responsiveness to hypoxia and hypercarbia (by anesthetics)
 - Respiratory **muscle fatigue** (neonates have a smaller % of type I fibers in their diaphragm and intercostal muscles)
 - **Hypothermia** and **anemia** can also contribute

CONT'D

- **Remedy** for POA or bradycardia
 - May be as simple as tactile stimulation
 - Sometimes mask ventilation (mechanical ventilation)
 - Those with life-threatening apnea and bradycardia before surgery may receive CNS stimulants
 - Caffeine and theophylline ↑ses central respiratory drive and ↓ses threshold for hypercarbia, stimulates diaphragmatic contractility

CONT'D

- Caffeine is favorable (wide margin and ↓sed toxicity)
- Its prophylactic administration to infants at risk of POA may prevent the need for prolonged postop ventilation
- Recommended loading dose is 10 mg/kg caffeine base
- Reginal anesthesia (without supplemental sedation) instead of general is beneficial

Which and for how long to monitor?

- Monitor all infants younger than 60 weeks postconceptual age overnight after surgery
- The incidence of significant POA and bradycardia is highest in the first 4 to 6 hours after surgery (it may occur after 12 hrs)

CONT'D

- Incidence of POA directly correlates to postconceptual age
- The younger the gestational age the higher the risk
- A meta-analysis study by Cote et al., determined that the risk of apnea could be correlated with a combination of gestational age and postconceptual age

Obstructive Sleep Apnea (OSA)

- Adenotonsillar hypertrophy with OSA is a frequent indication for tonsillectomy and adenoidectomy
- These patients are at risk for AWO preoperative sedation and during induction process
- They may exhibit worsening of their obstructive symptoms due to tissue edema, altered response to carbon dioxide, and residual effects of anesthetic agents postoperatively

CONT'D

- Children under 3-years of age are at particularly high risk for respiratory complications
- Children with severe OSA require postoperative observation in the hospital(for >4hrs)
- These issues should be discussed with family and the surgeon ahead of procedures

CONT'D

- OSA often accompanies obesity
- All patients with markedly elevated BMI should be asked about sleep apnea symptoms
- Obese children also have an increased incidence of difficult airway, upper airway obstruction in the PACU, extended PACU stays, and PONV

Asthma

- These children should be under optimal medical care prior to undergoing general anesthesia and surgery
- More active the disease (as indicated by recent asthma symptoms, asthma medication usage, and recent treatment in an ED), the greater probability of perioperative complications

CONT'D

- All oral and inhaled medications, such corticosteroids and β -agonists, should be continued up to the day of surgery
- Recent evidences show that use of inhaled short-acting β -agonists prior to induction of anesthesia eliminates the increase in airway pressure that is typically associated with intubation in asthmatic patients

Questions?

